

Liam Haynie

Salt Lake City, UT | 801-410-2257 | liamhaynie@gmail.com

EDUCATION

Bachelor of Science in Mechanical Engineering, Emphasis in Robotics & Control

University of Utah

May 2027

GPA: 3.49

- **Dean's List: Spring 2025 – Fall 2025, Summer 2026**

Associate of Science in Biology

Utah Valley University

2019 - 2021

TECHNICAL SKILLS

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|-----------------------------------|--|--------------------------------|
| • SolidWorks Certified, CSWA 2025 | • Digital multimeter (DMM) | • 3D printing (FDM) |
| • MATLAB | • Soldering (through hole & SMD) | • Statistical data analysis |
| • Arduino (C/C++) | • Hands-on prototyping experience | • Experimental data collection |
| • LTSpice | • Slicing software (Cura & Bambu Studio) | • Adobe Premiere |
| • Oscilloscope operation | | • Microsoft Office |
| | | • Documentation management |

EXPERIENCE

R&D Engineering Intern, *Medtronic Neurovascular*

Jul. 2026 – Sep. 2026

- Characterized unreleased neurovascular guidewires to establish production warning limits and investigate product performance issues, supporting R&D testing and quality decisions.
- Served as a key contributor during root cause analysis troubleshooting of neurovascular guidewire coating abnormalities. Developed DOE's, contributed to fish-bone diagrams and conducted testing to rectify coating issues.
- Solidworks designed and rapid-prototyped a universal mounting fixture for microscopic inspection of intravenous neurovascular devices, allowing for multi-axis rotational inspection and consistent digital analysis of devices as small as 300µm.
- Built an automated data processing workflow system for guidewire mechanical friction tests. Utilized MS Excel scripts and Power Automate to eliminate manual processing and expedite data analysis.

Solution Specialist (Infectious Disease Antigen Testing Lab), *ARUP Laboratories*

Oct. 2020 – Jul. 2026

- Collaborated with LabTech and ARUP R&D teams on the development of a *Streptococcus pneumoniae* ELISA by performing assay testing and supporting experimental validation.
- Supported research and development of a rapid *Escherichia coli* O157:H7 card assay by sourcing, de-identifying, and preparing previously tested clinical samples for Meridian Bioscience's Curian device.

Undergraduate Research Assistant (UROP), *Additive Manufacturing Lab, University of Utah*

Jan. 2022 – Aug. 2022

- Conducted sponsored undergraduate research in additive manufacturing focused on the development of bioabsorbable sensors for periodontal aligners and artificial joint applications.
- Supported experimental testing by capturing, organizing, and analyzing data from laboratory experiments to evaluate sensor performance.

PERSONAL PROJECT

Rat Operated Car | Personal Project (inspired by [Crawford et al., 2020](#))

Jun. 2025 – Dec. 2025

- Designed and built an original rat operated vehicle for emotional enrichment using laser cut and 3D printed components, featuring a rat centric ergonomic carriage and a three-pad copper touch interface with Arduino control for rat directed steering.
- Programmed Arduino firmware with touchpad debouncing, a state tracking feedback system, and ramped PWM acceleration/braking to smooth motion and reduce jitter.